Technical PRD

Data Scientist Assessment

Agriculture Database · MySQL · FastAPI · Pandas

1. Purpose & Goal

This document explains what a candidate needs to build for the Data Scientist assessment. Think of it as a clear instruction guide — not a technical spec full of jargon. Read it once and you should know exactly what to do.

> **The Goal in one sentence:** Connect to a MySQL agriculture database, process the data using Python and pandas, and serve it through a FastAPI web API with 8 endpoints across 2 reports.

2. What You Are Given

You will receive the following before starting the assessment:

Item	Details
MySQL Database	agriculture_db — pre-populated and ready to connect
Database Credentials	Host, port, username, and password will be provided separately
Data Dictionary PDF	Explains every table, column, data type, and example values
This PRD	Tells you what reports and endpoints to build

The database has 6 tables in a Star Schema design:

Table	What it contains	Rows
fact_harvest_sales	Core data — every harvest and sale event	220+
dim_farm	Farm details: name, owner, region, size	30
dim_crop	Crop types, categories, growing seasons	20
dim_date	Dates with year, quarter, season attributes	25
dim_input_supply	Fertilizers, seeds, pesticides used	20
dim_market	Market channels and buyer info	20

3. Technology Stack

You must use the following tools. No substitutions.

Layer	Tool / Library
Database	MySQL (already running — you just connect to it)
DB Connection	SQLAlchemy + PyMySQL

Data Processing	Python + pandas
API Framework	FastAPI
API Server	Uvicorn
Python Version	3.9 or above

Quick connection snippet:

```
from sqlalchemy import create_engine
import pandas as pd

engine = create_engine('mysql+pymysql://user:password@host:3306/agriculture_db')
df = pd.read_sql('SELECT * FROM vw_harvest_full', engine)
```

4. Report 1 — Farm Performance Report

This report answers the question: how well are our farms doing? It looks at revenue, profit, losses, and rankings across different farms, regions, and time periods.

Endpoint 1 — Farm Summary

GET	`/farms/summary`

What it does: Returns a summary of all farms — total revenue, total profit, total cost, and average loss percentage. You can narrow down the results using the filters below.

Query Filters

`region=`	`farm_type=`	`year=`	`season=`

Example JSON Response


```
{
  "total_farms": 12,
  "filters_applied": {
    "region": "Dhaka",
    "year": 2023
  },
  "data": [
    {
      "farm_name": "Green Valley Farm",
      "region": "Dhaka",
      "farm_type": "Large",
      "total_revenue_bdt": 4250000,
      "total_cost_bdt": 980000,
      "net_profit_bdt": 3270000,
      "avg_loss_pct": 4.2
    },
```

```
    {
      "farm_name": "Golden Harvest",
      "region": "Dhaka",
      "farm_type": "Small",
      "total_revenue_bdt": 1120000,
      "total_cost_bdt": 420000,
      "net_profit_bdt": 700000,
      "avg_loss_pct": 6.8
    }
  ]
}
```


💡 **Note:** If no filters are passed, it returns data for ALL farms across all years.

Endpoint 2 — Single Farm Performance

GET	`/farms/{farm_id}/performance`

What it does: Returns a detailed breakdown for one specific farm. Shows revenue and profit per crop, per year, and per market channel — all filterable.

Query Filters

year=	crop_category=	market_type=

Example JSON Response


```
{
  "farm_id": 1,
  "farm_name": "Green Valley Farm",
  "owner": "Rahman Ali",
  "region": "Dhaka",
  "filters_applied": {
    "year": 2023,
    "crop_category": "Cereal"
  },
  "performance": [
    {
      "crop_name": "Boro Rice",
      "year": 2023,
      "market_type": "Wholesale",
      "quantity_sold_ton": 60.0,
      "revenue_bdt": 1560000,
      "net_profit_bdt": 1180000,
      "quality_grade": "A"
    }
  ]
}
```


💡 **Note:** farm_id is an integer from 1 to 30. You can find all farm IDs in dim_farm.

Endpoint 3 — Top Farms Ranking

GET	`/farms/top`

What it does: Returns the top N farms ranked by a chosen metric. You can choose to rank by profit, revenue, or yield efficiency. Filter by region, farm type, and year.

Query Filters

`metric=profit|revenue|yield`	`region=`	`farm_type=`	`year=`	`limit=`

Example JSON Response


```json
{
  "metric": "profit",
  "filters_applied": {
    "region": "Rajshahi",
    "farm_type": "Commercial",
    "limit": 5
  },
  "rankings": [
    {
      "rank": 1,
      "farm_name": "Barind Farms",
      "region": "Rajshahi",
      "farm_type": "Commercial",
      "net_profit_bdt": 8900000,
      "total_revenue_bdt": 11200000
    },
    {
      "rank": 2,
      "farm_name": "Padma Shore",
      "region": "Rajshahi",
      "farm_type": "Large",
      "net_profit_bdt": 5400000,
      "total_revenue_bdt": 7100000
    }
  ]
}
```


> 💡 **Note:** Default limit is 10 if not specified. Default metric is profit.

Endpoint 4 — Loss Analysis

GET	`/farms/loss-analysis`

What it does: Shows post-harvest loss data — how much produce was lost (in tonnes and percentage) broken down by region, season, crop, and quality grade. Helps identify problem areas.

Query Filters

region=	year=	season=	quality_grade=	crop_category=

Example JSON Response


```
{
  "filters_applied": {
    "season": "Kharif",
    "year": 2023,
    "quality_grade": "C"
  },
  "summary": {
    "total_harvested_ton": 1240.5,
    "total_lost_ton": 98.2,
    "overall_loss_pct": 7.9
  },
  "breakdown": [
    {
      "region": "Chittagong",
      "crop_category": "Cereal",
      "quality_grade": "C",
      "total_lost_ton": 42.5,
      "loss_pct": 9.1,
      "pesticide_residue": "Trace"
    },
    {
      "region": "Sylhet",
      "crop_category": "Spice",
      "quality_grade": "C",
      "total_lost_ton": 55.7,
      "loss_pct": 11.3,
      "pesticide_residue": "Low"
    }
  ]
}
```


💡 **Note:** This endpoint is great for finding which regions and seasons have the worst harvest losses.

5. Report 2 — Crop & Market Intelligence Report

This report answers the question: which crops and markets are performing best? It looks at yield efficiency, seasonal trends, pricing, and quality distribution across different crops and market channels.

Endpoint 5 — Crop Yield Efficiency

GET	`/crops/yield-efficiency`

What it does: Compares the actual yield per hectare achieved on our farms against the national benchmark yield for each crop. Shows how efficient each crop's farming is.

Query Filters

crop_category=	season=	year=	region=	water_requirement=

Example JSON Response


```json
{
  "filters_applied": {
    "crop_category": "Cereal",
    "year": 2023
  },
  "data": [
    {
      "crop_name": "Boro Rice",
      "crop_category": "Cereal",
      "avg_yield_benchmark_ton_per_ha": 5.2,
      "actual_avg_yield_ton_per_ha": 5.8,
      "efficiency_pct": 111.5,
      "total_area_planted_ha": 186.0,
      "season": "Rabi"
    },
    {
      "crop_name": "Wheat",
      "crop_category": "Cereal",
      "avg_yield_benchmark_ton_per_ha": 3.1,
      "actual_avg_yield_ton_per_ha": 2.8,
      "efficiency_pct": 90.3,
      "total_area_planted_ha": 95.0,
      "season": "Rabi"
    }
  ]
}
```


> 💡 **Note:** efficiency_pct above 100 means the farm is outperforming the national average. Below 100 means underperforming.

Endpoint 6 — Seasonal Revenue Trend

GET `/crops/seasonal-trend`

What it does: Shows how revenue and quantity sold changes across different seasons and years for each crop. Useful for spotting which seasons are most profitable for which crops.

Query Filters

crop_name=	crop_category=	year=	quarter=	market_type=

Example JSON Response


```
{
  "filters_applied": {
    "crop_category": "Vegetable",
    "year": 2023
  },
  "trend": [
    {
      "crop_name": "Potato",
      "year": 2023,
      "quarter": 1,
      "season": "Winter",
      "total_quantity_sold_ton": 392.0,
      "total_revenue_bdt": 8624000,
      "avg_price_per_ton_bdt": 22000,
      "num_harvests": 4
    },
    {
      "crop_name": "Tomato",
      "year": 2023,
      "quarter": 1,
      "season": "Winter",
      "total_quantity_sold_ton": 228.0,
      "total_revenue_bdt": 7296000,
      "avg_price_per_ton_bdt": 32000,
      "num_harvests": 3
    }
  ]
}
```


> 💡 **Note:** quarter filter accepts values 1, 2, 3, or 4. If omitted, all quarters are returned.

Endpoint 7 — Market Price Comparison

GET `/markets/price-comparison`

What it does: Compares average selling prices across different market types, districts, and price tiers for each crop. Helps identify which market channel gives the best return.

Query Filters

market_type=	crop_category=	year=	season=	price_tier=	district=

Example JSON Response


```
{
  "filters_applied": {
    "crop_category": "Cereal",
    "year": 2023,
    "market_type": "Export"
  },
  "comparison": [
    {
      "market_name": "Chittagong Port Market",
      "market_type": "Export",
      "price_tier": "Premium",
      "district": "Chittagong",
      "crop_name": "Aman Rice",
      "avg_price_per_ton_bdt": 48000,
      "total_quantity_sold_ton": 87.0,
      "total_revenue_bdt": 4176000
    },
    {
      "market_name": "Narayanganj Port",
      "market_type": "Export",
      "price_tier": "Premium",
      "district": "Narayanganj",
      "crop_name": "Boro Rice",
      "avg_price_per_ton_bdt": 52000,
      "total_quantity_sold_ton": 60.0,
      "total_revenue_bdt": 3120000
    }
  ]
}
```


> 💡 **Note:** Export markets consistently return Premium prices. Compare against Local market prices to see the difference.

Endpoint 8 — Quality Grade Breakdown

GET	`/crops/quality-breakdown`

What it does: Shows the distribution of quality grades (A, B, C, D) for crops across different regions, markets, and years. Also shows pesticide residue levels alongside quality.

Query Filters

crop_id=	crop_category=	year=	region=	market_type=	pesticide_residue=

Example JSON Response


```
{
```

```
  "filters_applied": {
    "crop_category": "Fruit",
    "year": 2023,
    "region": "Rajshahi"
  },
  "total_records": 8,
  "grade_distribution": {
    "A": { "count": 5, "pct": 62.5, "avg_revenue_bdt": 1840000 },
    "B": { "count": 2, "pct": 25.0, "avg_revenue_bdt": 920000 },
    "C": { "count": 1, "pct": 12.5, "avg_revenue_bdt": 380000 },
    "D": { "count": 0, "pct": 0.0,  "avg_revenue_bdt": 0 }
  },
  "pesticide_residue_breakdown": {
    "None":  { "count": 6, "pct": 75.0 },
    "Trace": { "count": 2, "pct": 25.0 },
    "Low":   { "count": 0, "pct": 0.0 },
    "High":  { "count": 0, "pct": 0.0 }
  }
}
```


💡 **Note:** Grade A produce commands the highest market price. Use this endpoint to correlate quality with pesticide residue levels.

6. Filter Reference — All Accepted Values

Every filter parameter and its accepted values are listed below. Passing an invalid value should return a clear error message from the API.

Filter Parameter	Accepted Values
region	Dhaka, Chittagong, Sylhet, Rajshahi, Khulna, Rangpur, Barisal, Mymensingh
farm_type	Small, Medium, Large, Commercial
crop_category	Cereal, Vegetable, Fruit, Pulse, Oilseed, Cash Crop, Spice
season	Spring, Summer, Autumn, Winter
growing_season	Rabi, Kharif, Zaid, Year-Round
market_type	Local, Wholesale, Export, Retail, Government Procurement
price_tier	Low, Medium, High, Premium
quality_grade	A, B, C, D
pesticide_residue	None, Trace, Low, High
water_requirement	Low, Medium, High
year	2022, 2023, 2024
quarter	1, 2, 3, 4
metric	profit, revenue, yield
limit	Any positive integer. Default: 10

7. What You Must Submit

When you are done, submit the following:

A Public Github Repo Including

- all files related to the projects
- requirements.txt is must
- README.md (Must include how to run)
- Dockerfile (for bonus point)

How we will test it: We will run your FastAPI server locally, hit each endpoint with different filter combinations, and verify the JSON responses are correct and complete.

8. How You Will Be Evaluated

Your submission will be scored across these 5 areas:

Area	What We Look For	Weight
Database Connection	Clean SQLAlchemy setup, no hardcoded credentials	15%
pandas Processing	Correct aggregations, filtering logic, efficient queries	30%
FastAPI Endpoints	All 8 endpoints working, correct filters, correct JSON shape	30%
Code Quality	Readable code, proper naming, no duplicate logic	15%
Error Handling	Graceful errors for invalid filters or missing data	10%
Docker File (Bonus)	A Proper Docker file/ Docker Compose file	10% of Total Score

9. Helpful Tips

- Start by connecting to the database and running SELECT * FROM vw_harvest_full to understand the full data shape.
- Use the 3 pre-built views (vw_harvest_full, vw_revenue_by_crop_year, vw_farm_profitability) — they save a lot of query writing.
- Build one endpoint at a time. Test it fully before moving to the next.
- Use pandas .query() or boolean indexing to apply filters cleanly after loading data.
- Return HTTP 422 with a clear message if a filter value is invalid (e.g. region='InvalidCity').
- Use FastAPI's Query() with description= to document your filter parameters — it makes the auto-generated docs at /docs much cleaner.
- revenue_bdt and net_profit_bdt are pre-computed in the database — you do not need to calculate them yourself.